SM2501/FSG3084

Lecture 8

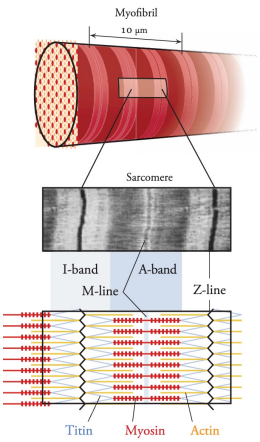
Muscle biomechanics and modelling, part 2

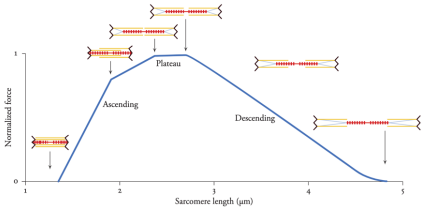


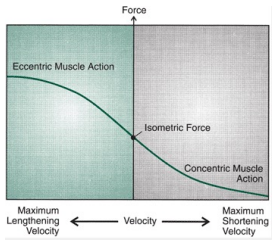


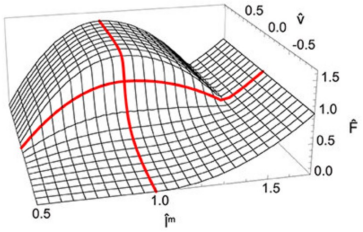
1

Review from Tuesday: Muscle biomechanics



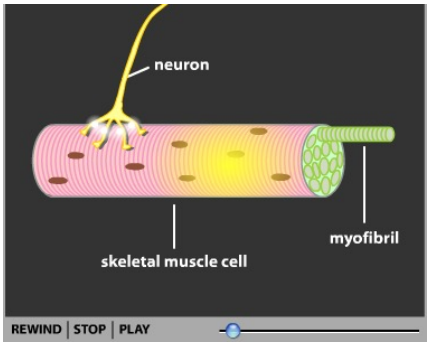






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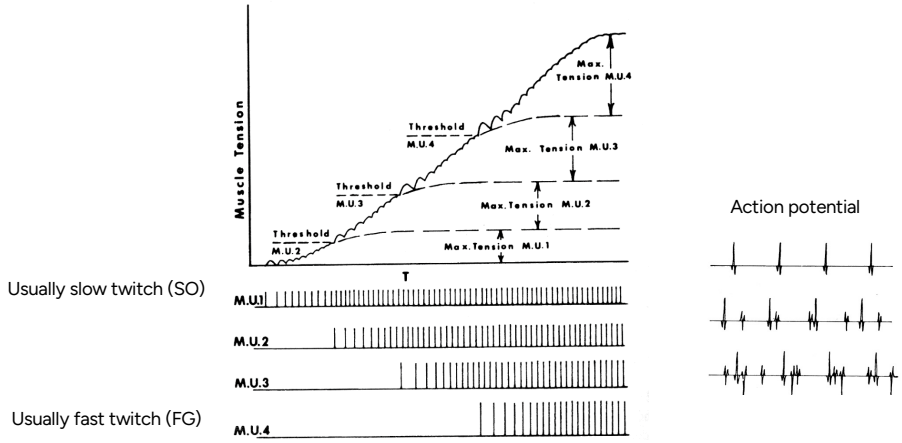
Activation and Sliding Filament Theory



http://www.sumanasinc.com/webcontent/animations/content/muscle.html
Garland Science Publishing

3

Recruitment of motor units



Winter 2005

4

Motor unit size

The number of muscle fibers in a motor unit varies among muscles.

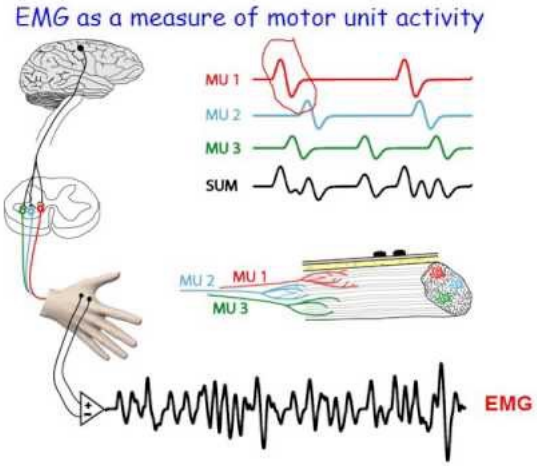
- In the hand and eye, fewer than **100 muscle fibers** can occupy a motor unit.
- In the lower leg, a single motor unit may contain as many as **1,000 muscle fibers**

Muscles which possess more motor units (and thus have greater individual motor neuron innervation) are able to control force output more finely.

(Buchthal & Schmalbruch, 1980)

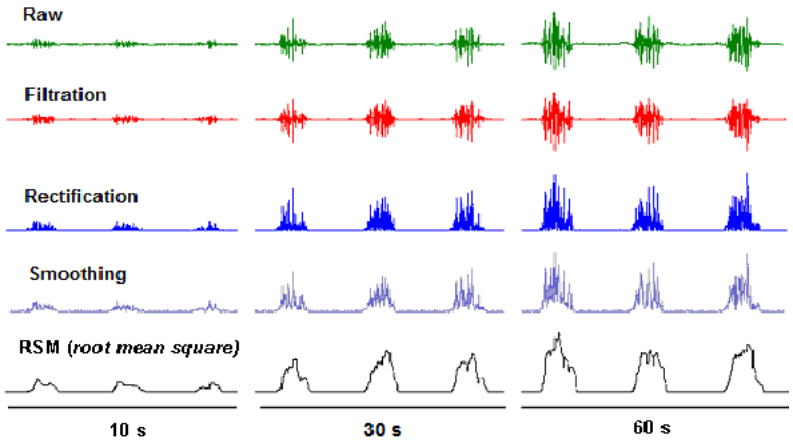
5

Electromyography (EMG)



6

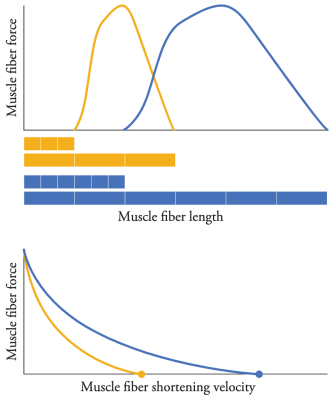
EMG Signal Processing



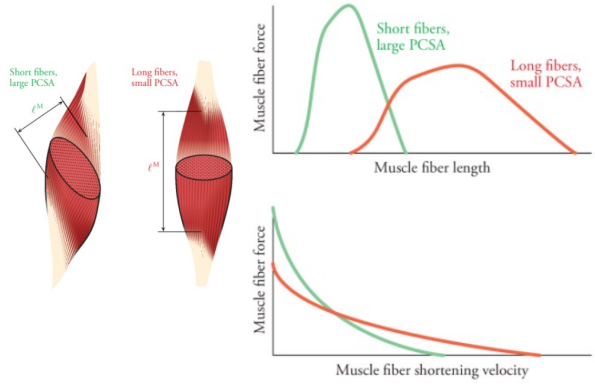
7

Sarcomere organization and fiber force

Muscle fiber force



Sarcomere distribution and fiber force

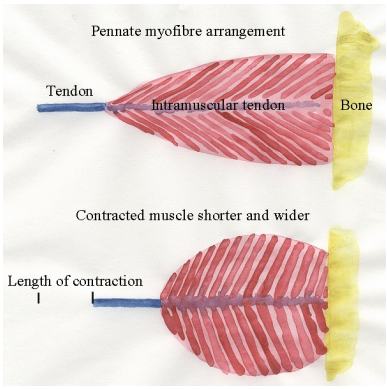


...All else being equal, a pennated muscle will be able to generate greater force than a parallel-fibered muscle of the same volume, but over a smaller range of lengths and at lower contraction velocities. -S. Delp

Uchida & Delp 2020

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Pennation angle is generally not constant



.....but usually modelled as fixed

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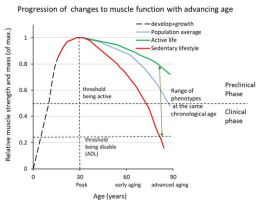
Muscles and exercise

- Resistance training leads to muscle hypertrophy fiber diameter increase.
  - Can increase number of myofibrils and development of connective tissue
  - Can increase amount of Type IIx fibers
- Aerobic and endurance exercise can lead to
  - Increased capillarization; improved oxygen & nutrition delivery
  - More fatigue-resistant muscle
  - Hypertrophy; higher muscle diameter, shift to Type I fibers
- High-intensity training can lead to more Type IIa fibers
- Exercise can stimulate neuron growth; strengthens neural pathways that control and coordinate movement
-

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Muscles and aging

- Muscle mass highest around age 30



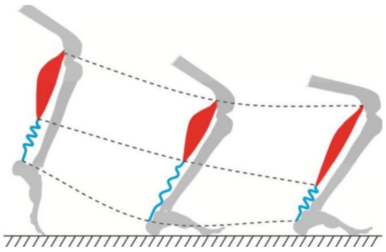
Gustafsson & Ulfhake, Int J Mol Sci 2024

- Adults 30-~50 lose ~3-8% muscle mass per decade
  - Loss accelerates at around age 50 for both men and women
  - At age 80, women have lost around 50% strength in back muscles. Men have lost around 64%.
- The number and size of Type II fibers reduces with age
- Muscle is fully capable of responding to resistance training.
  - In a study with adults over 90 (Frontera et al, JAMA 1990), with progressive resistance training
    - After 8 weeks, thigh muscle area increased 10%, equivalent to ~1 decade of loss
    - Strength increased by 175% after the program

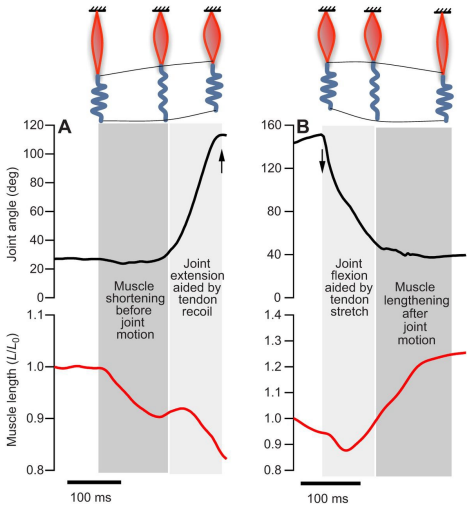
11

11/13/25

Muscle-*tendon* units (MTUs) length changes
(or muscle-tendon actuators)



...Often overlooked in models



Roberts TJ, J Exp Biol 2016 219;266-75

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Real MTUs

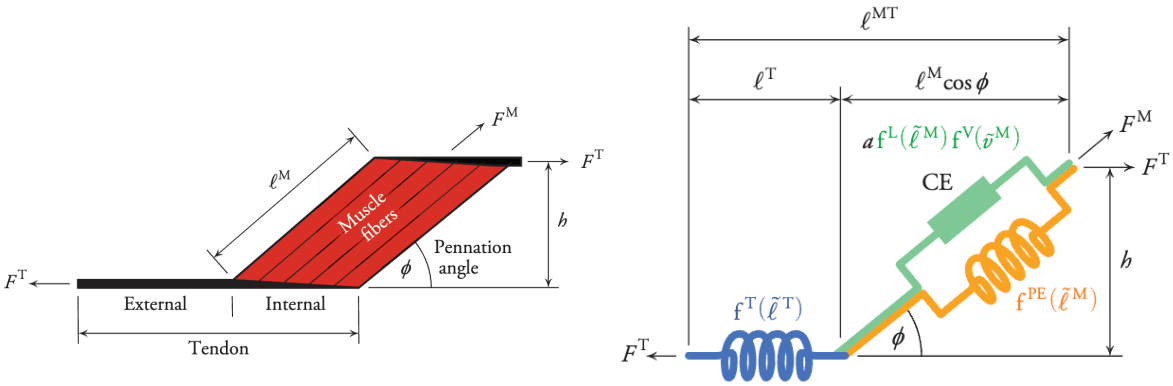
- Function determined by fiber types, attachment and volume geometry geometry, tissue properties, fiber diameters, muscle mass, metabolic status, anatomical variation, tendon length and compliance, pain, subjective motivation, etc. etc.

Modelled MTUs

- Function and structure described by a model with a few variable parameters.

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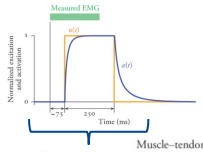
Hill-type model of muscle-tendon dynamics



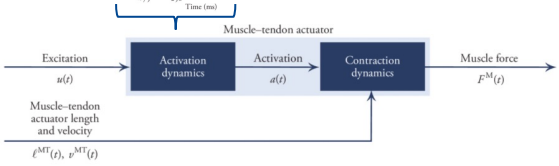
Uchida & Delp 2020

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Excitation -> MTU model force: putting it together

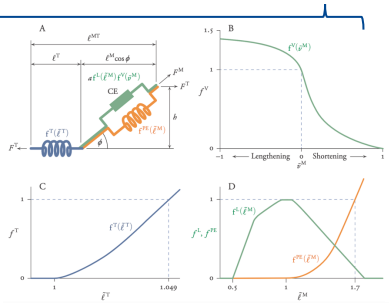


Excitation dynamics described by activation and deactivation constants τ_A and τ_D



$$l^{MT} = l^M \cos\phi + l^T$$

$$l^M \cos\phi \propto a f^L(\tilde{l}^M)\, f^V(\tilde{v}^M)$$



$$F^M(t) = F_o^M\left[a(t)\, f^L\left(\tilde{\ell}^M(t)\right) f^V\left(\tilde{v}^M(t)\right) + f^{PE}\left(\tilde{\ell}^M(t)\right)\right]$$

Uchida & Delp 2020

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Scale: multibody system

Moment arm and MTU length determined by kinematics

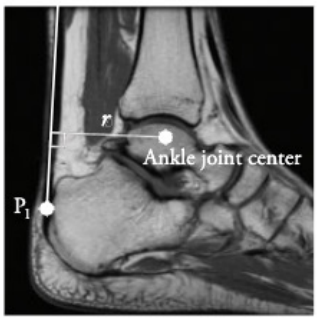
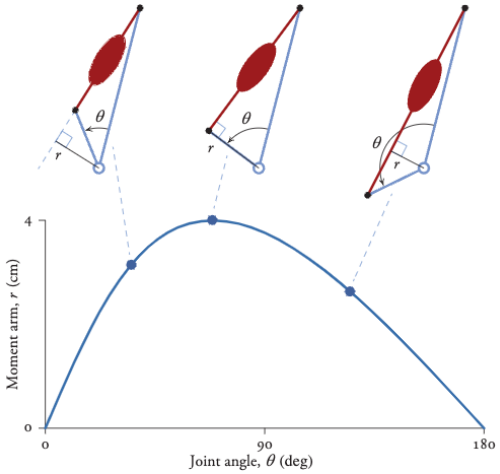


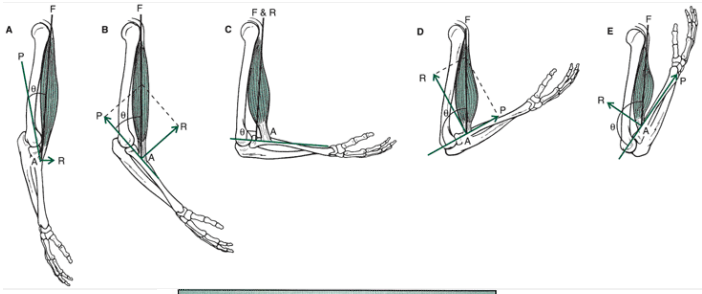
FIGURE 6.3
Estimating muscle moment arm from a magnetic resonance image. MRI courtesy of Alex Frietas.

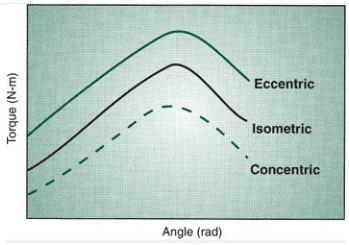


Delp & Uchida 2020

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Muscles' moment generation throughout range of motion





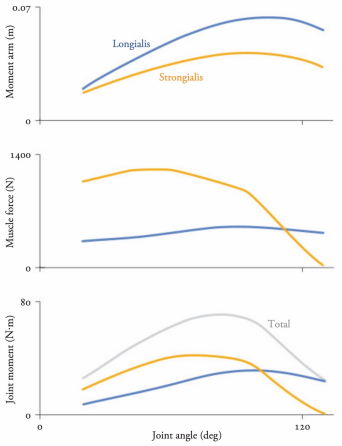
Lengthening

Shortening

Hamill & Knutzen 2009

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Peak joint moment



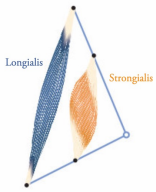
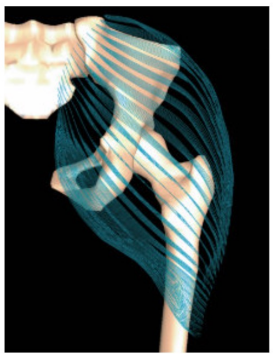


FIGURE 6.13
The peak moment generated by a muscle occurs at a joint angle that corresponds to neither peak moment arm nor peak force, demonstrated here for two fictitious muscles. The total joint moment generated by a muscle group is simply the sum of the moments generated by all muscles in the group.

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Muscle volume, attachments and geometry



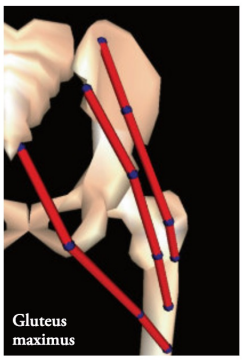
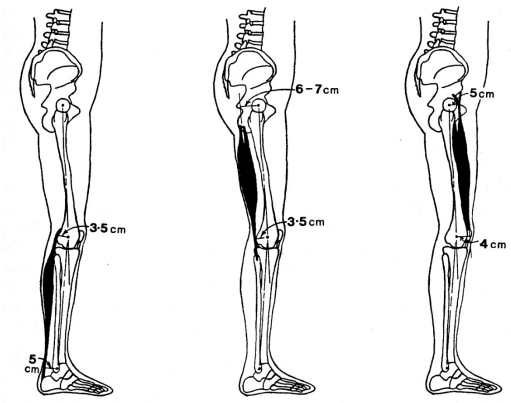


FIGURE 5.13

Commonly modelled a finite number of line elements & wrapping points

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Biarticular muscles



Winter 2004

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